

GNR638: Coding Assignment-2

Pre-trained CNN Representation Transfer and Robustness Analysis

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1 Introduction

Pre-trained convolutional neural networks (CNNs) trained on large-scale datasets such as ImageNet have become de-facto feature extractors for downstream vision tasks. This assignment systematically investigates the behavior of three selected CNN backbones on the **Aerial Images Dataset (AID)** containing 30 land-use classes.

The key research questions addressed are:

- Q1.** Which architecture transfers best and why?
- Q2.** Which model is most robust under distribution shift?
- Q3.** Which layers encode the most transferable semantics?
- Q4.** When does fine-tuning degrade robustness?
- Q5.** How does inductive bias manifest empirically?

2 Experimental Setup

2.1 Selected Backbones

We selected three ImageNet-pre-trained CNN backbones using the `timm` library:

Table 1: Selected Pre-trained CNN Backbones

Model	Params (M)	MACs (G)	FLOPs (G)
ResNet-50	23.57	4.13	4.11
DenseNet-121	6.90	2.83	2.86
EfficientNet-B0	4.05	0.38	0.40

2.2 Dataset

The **Aerial Images Dataset (AID)** consists of aerial scene images across 30 categories. The dataset is organized in an image-folder format with JPEG images.

Table 2: Dataset Statistics

Property	Value
Number of Classes	30
Total Images	~6,993
Train / Val Split	80% / 20%
Train Samples	5,594
Val Samples	1,399
Image Resolution (resize)	224×224
Random Seed	42

2.3 Training Configuration

Table 3: Common Hyperparameters Across All Scenarios

Hyperparameter	Value
Optimizer	Adam
Learning Rate	1×10^{-3}
Weight Decay	1×10^{-4}
Batch Size	64
Max Epochs (full)	30
Max Epochs (few-shot)	20
LR Scheduler	None
Image Size	224×224
Train / Val Split	80% / 20%
Train Augmentation	RandomHorizontalFlip, RandomVerticalFlip, ColorJitter (brightness=0.2, contrast=0.2)
Val Augmentation	None (resize + normalize only)
Normalization	mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]
Loss Function	CrossEntropyLoss
Random Seed	42

2.4 Efficiency Metrics

Model efficiency is measured using:

- **Parameters:** Total trainable and non-trainable weights
- **MACs:** Multiply-Accumulate Operations per forward pass
- **FLOPs:** Computed via `fvcore` library

These metrics are computed for input size $1 \times 3 \times 224 \times 224$.

Scenario 1: Linear Probe Transfer

Methodology

All backbone parameters are **frozen**. Only a single linear classification head ($d_{\text{feat}} \rightarrow 30$) is trained. This isolates the quality of pre-trained representations without any adaptation. Under the linear probe setting, the trainable parameter fractions are extremely small: 0.26% for ResNet-50 (0.0615 M), 0.44% for DenseNet-121 (0.0307 M), and 11.07% for EfficientNet-B0 (0.4480 M).

Quantitative Results

Table 4: Linear Probe Transfer — Final Validation Accuracy

Model	Train Acc (%)	Best Val Acc (%)	Final Val Acc (%)	Train-Val Gap	Feature Dim
ResNet-50	92.89	87.63	87.63	5.25	2048
DenseNet-121	97.94	90.85	89.85	8.09	1024
EfficientNet-B0	99.16	91.14	89.56	9.60	1280

Training and Validation Accuracy Curves

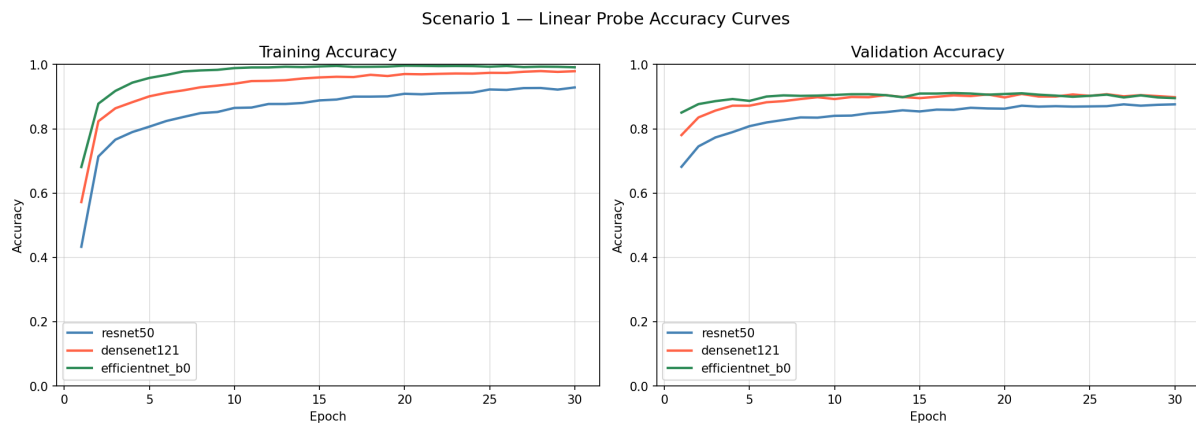


Figure 1: Linear Probe: Training and Validation Accuracy Curves (Epoch 1–30). ResNet-50 converges steadily to 87.6% validation accuracy, while DenseNet-121 and EfficientNet-B0 plateau near 91% but show slightly larger training-validation gaps, indicating mild overfitting in the linear classifier head.

Confusion Matrix

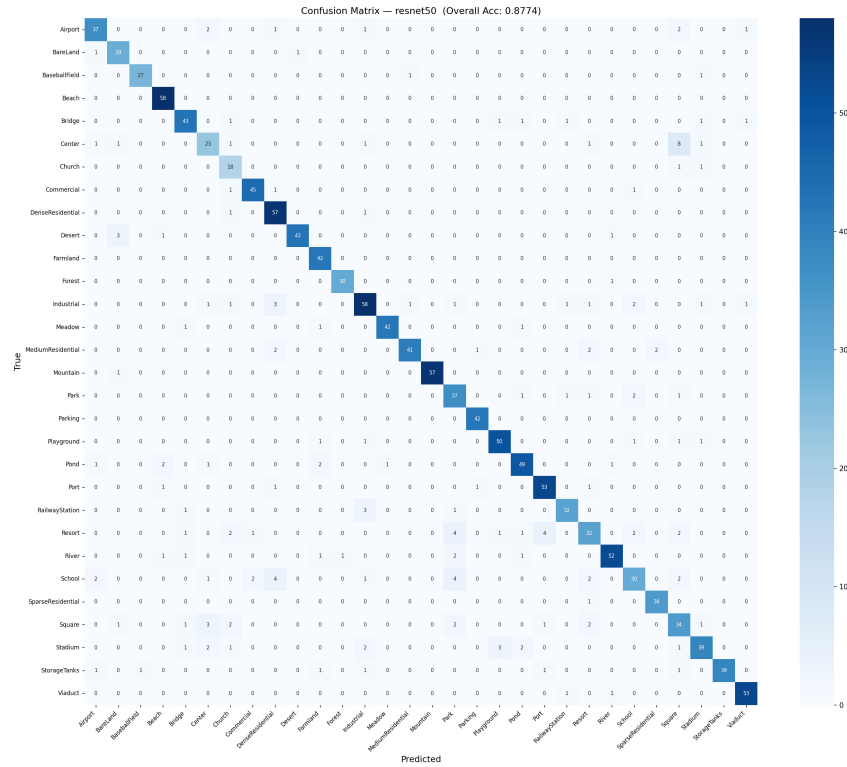


Figure 2: Confusion matrix for ResNet-50 (Linear Probe).

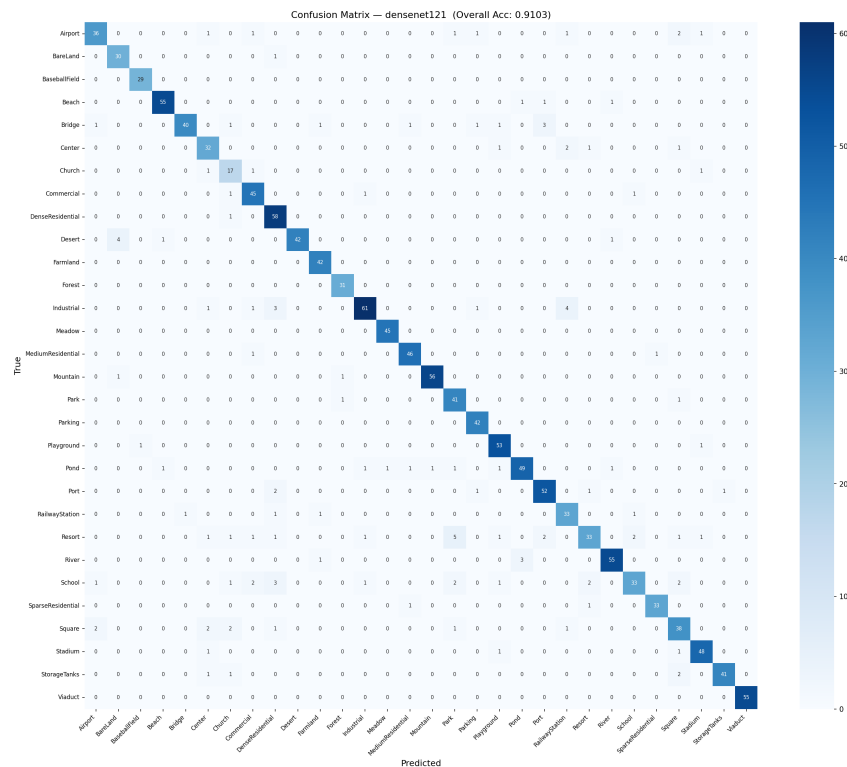


Figure 3: Confusion matrix for DenseNet-121 (Linear Probe).

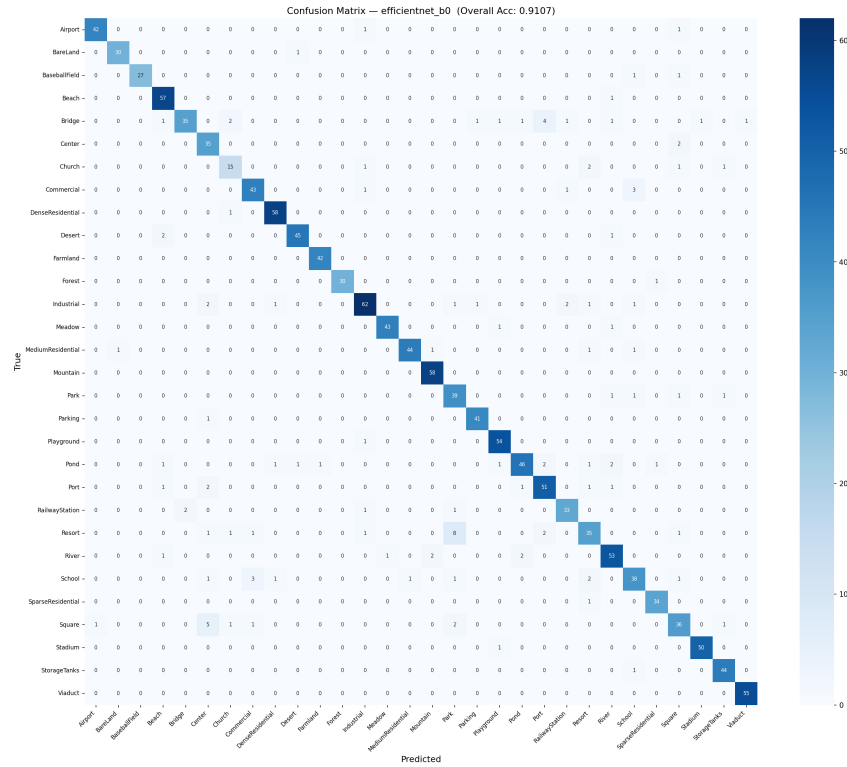


Figure 4: Confusion matrix for EfficientNet-B0 (Linear Probe).

Feature Embedding Visualization (PCA + t-SNE)

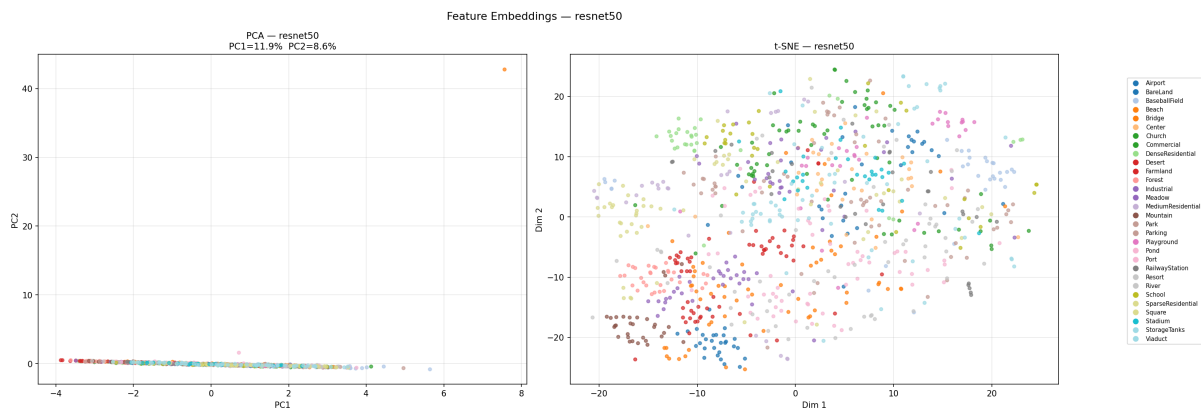


Figure 5: ResNet-50

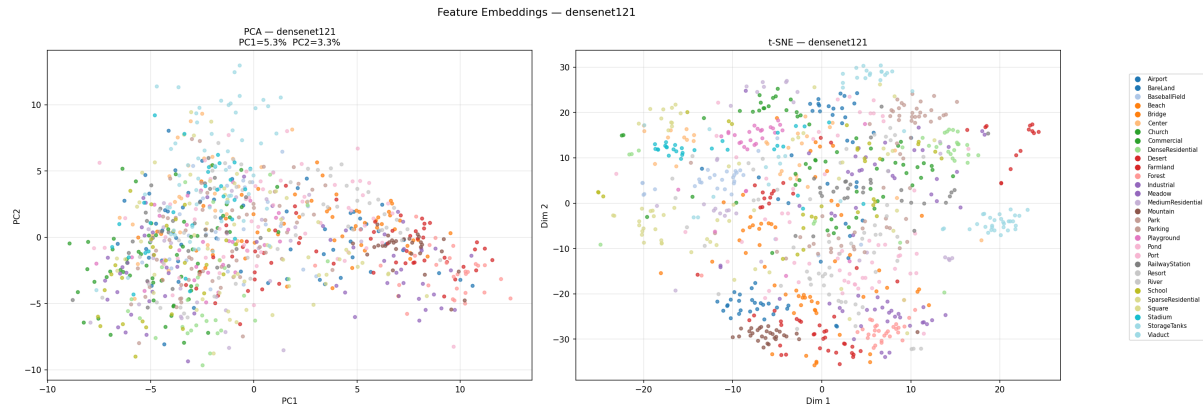


Figure 6: DenseNet-121

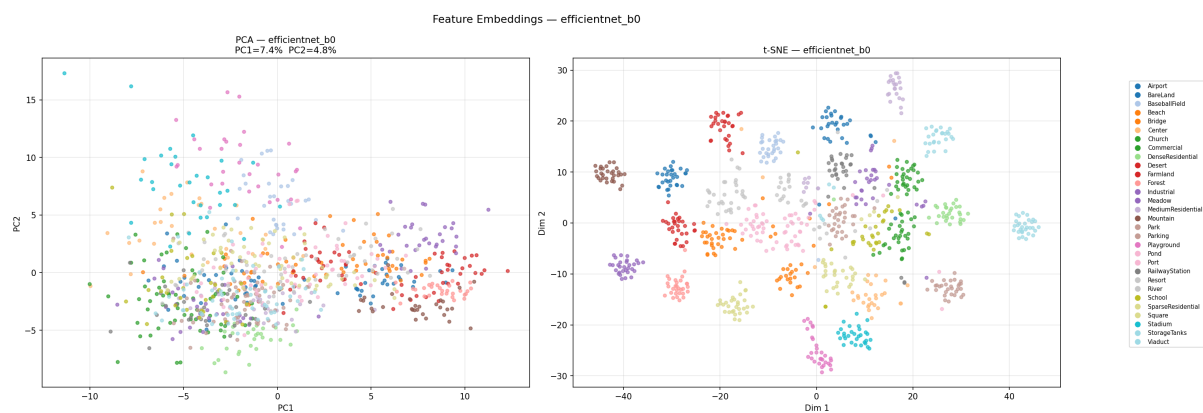


Figure 7: EfficientNet-B0

Feature Separability Metrics

Table 5: Feature Separability Metrics (900-sample fixed subset)

Model	Silhouette Score	Mean Inter-class Dist	Mean Intra-class Dist	Separability Ratio
ResNet-50	-0.0308	2.7670	5.9593	0.4643
DenseNet-121	0.0131	12.2253	25.7860	0.4741
EfficientNet-B0	0.0985	15.1785	18.7611	0.8090

Table 6: Computational Budget — Scenario 1 (Linear Probe)

Model	Epochs	Time (min)	Device	Batch	LR	FLOPs (G)	MACs (G)	Trainable (%)
ResNet-50	30	45.61	cuda	64	1e-3	4.11	4.13	0.26
DenseNet-121	30	47.71	cuda	64	1e-3	2.86	2.83	0.44
EfficientNet-B0	30	45.13	cuda	64	1e-3	0.40	0.38	11.07
Total	90	138.45						

Analysis

Feature Separability. EfficientNet-B0 achieves the highest silhouette score (0.0985) and the best separability ratio (0.8090), meaning its inter-class distances are large relative to intra-class spread despite its compact 4.05 M parameter footprint. ResNet-50 yields a *negative* silhouette score (-0.0308), indicating that its 2048-dimensional features are less discriminatively organized for aerial scene categories under a linear probe. DenseNet-121 achieves an intermediate separability ratio of 0.4741, consistent with its dense feature reuse architecture.

Transfer Quality. All three backbones transfer meaningfully without any fine-tuning: best validation accuracies of 87.63% (ResNet-50), 90.85% (DenseNet-121), and 91.14% (EfficientNet-B0) demonstrate that ImageNet representations generalize to aerial imagery. EfficientNet-B0 achieves the highest best validation accuracy despite being the smallest model (4.05 M parameters).

Overfitting Analysis. EfficientNet-B0 shows the largest train-val gap (9.60%) under the linear probe setting. The 1280-dimensional feature space provides high capacity for the linear classifier to memorize the 5,594 training samples. DenseNet-121 shows an 8.09% gap, while ResNet-50’s 2048-dimensional features result in the smallest gap (5.25%).

Failure Cases. The primary failure cases are consistent across all three backbones. For ResNet-50, the worst-performing classes are *Center* (62.16%), *School* (62.50%), and *Resort* (64.00%). For DenseNet-121, failures concentrate on *Resort* (66.00%), *School* (68.75%), and *Square* (80.85%). For EfficientNet-B0, the hardest classes are *Resort* (70.00%), *Bridge* (71.43%), and *Church* (75.00%).

Scenario 2: Fine-Tuning Strategies

Methodology

Four fine-tuning strategies are compared under identical hyperparameters (Adam, $\text{lr}=1 \times 10^{-3}$, weight decay= 1×10^{-4} , batch size=64, 30 epochs):

1. **Linear Probe:** Only classification head trained (backbone fully frozen).
2. **Last Block Fine-Tuning:** Final residual/dense/MBConv block + head unfrozen.
3. **Full Fine-Tuning:** Entire backbone + head trained.
4. **Selective Unfreezing ($\leq 20\%$):** Automated strategy described below.

Selective Unfreezing Strategy (Bonus Part 2 — Automated). Instead of manually choosing layers, we implemented an automated parameter selection algorithm. All backbone parameters are first frozen. Starting from the *deepest* layers, parameters are progressively unfrozen until the cumulative trainable parameter count reaches approximately 20% of total model parameters.

Algorithm 1 Automated Selective Unfreezing

```

1: Freeze all parameters:  $\forall p \in \theta, p.requires\_grad \leftarrow \text{False}$ 
2: trainable  $\leftarrow 0$ , total  $\leftarrow \sum_p |p|$ 
3: for (name, param) in REVERSED(model.named_parameters()) do
4:   param.requires_grad  $\leftarrow \text{True}$ 
5:   trainable  $\leftarrow$  trainable +  $|param|$ 
6:   if trainable/total  $\geq 0.20$  then
7:     break
8:   end if
9: end for

```

Percentage of Unfrozen Parameters

Table 7: Trainable Parameters Per Strategy Per Model

Strategy	ResNet-50		DenseNet-121		EfficientNet-B0	
	Params	%	Params	%	Params	%
Linear Probe	61,470	0.26	30,750	0.44	448,030	11.07
Last Block	15,026,206	63.75	2,188,830	31.34	1,165,262	28.80
Selective ($\leq 20\%$)	5,576,734	23.66	1,414,622	20.25	819,870	20.26
Full Fine-Tuning	23,569,502	100.00	6,984,606	100.00	4,045,978	100.00

Best Validation Accuracy Across Strategies

Table 8: Best Validation Accuracy (%) Per Strategy Per Model

Model	Linear Probe	Last Block	Selective ($\leq 20\%$)	Full Fine-Tune
ResNet-50	88.46	95.02	91.80	94.59
DenseNet-121	91.45	95.02	94.74	94.12
EfficientNet-B0	91.14	93.43	92.58	95.29

Accuracy vs. Percentage of Unfrozen Parameters

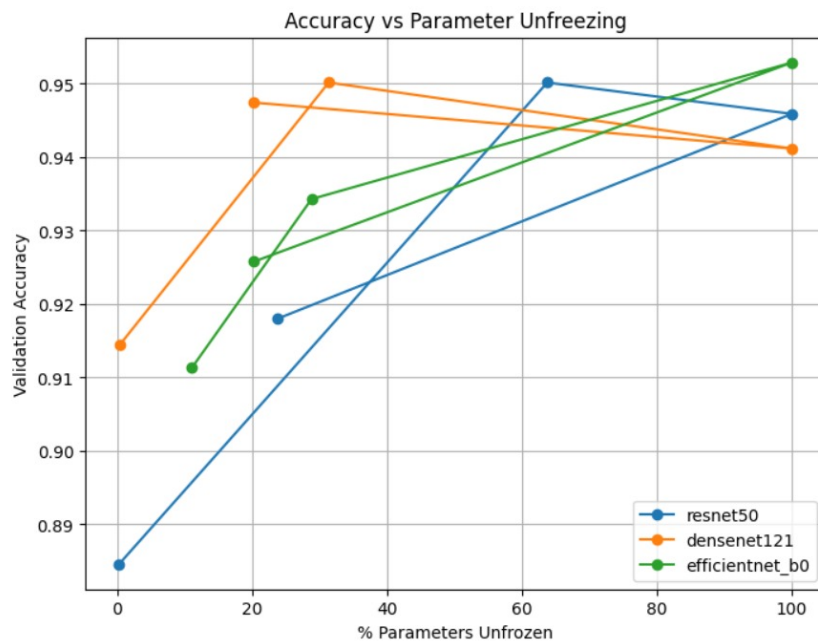


Figure 8: Validation Accuracy vs. Percentage of Unfrozen Parameters.

Validation Accuracy Across Strategies

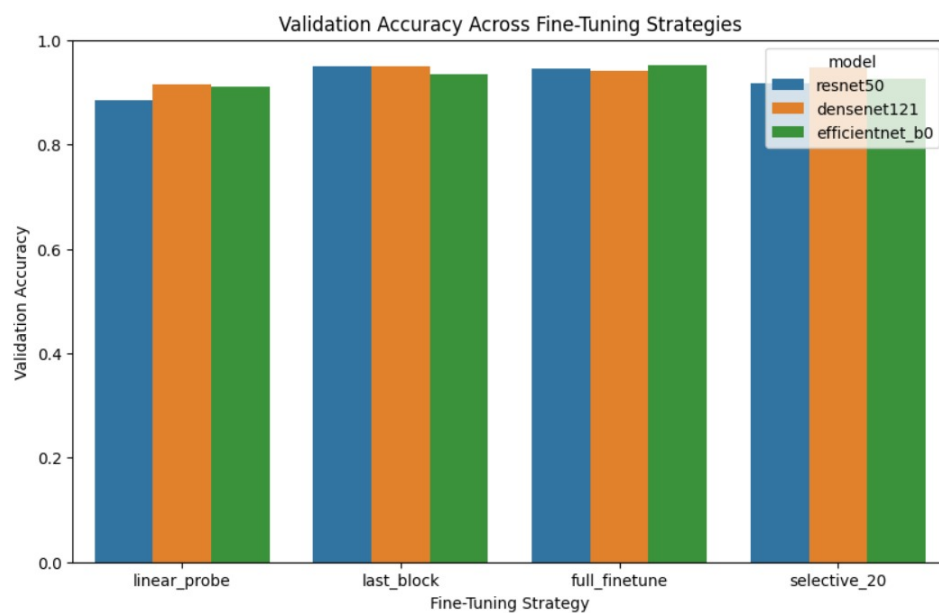


Figure 9: Grouped bar chart of best validation accuracy across all four fine-tuning strategies.

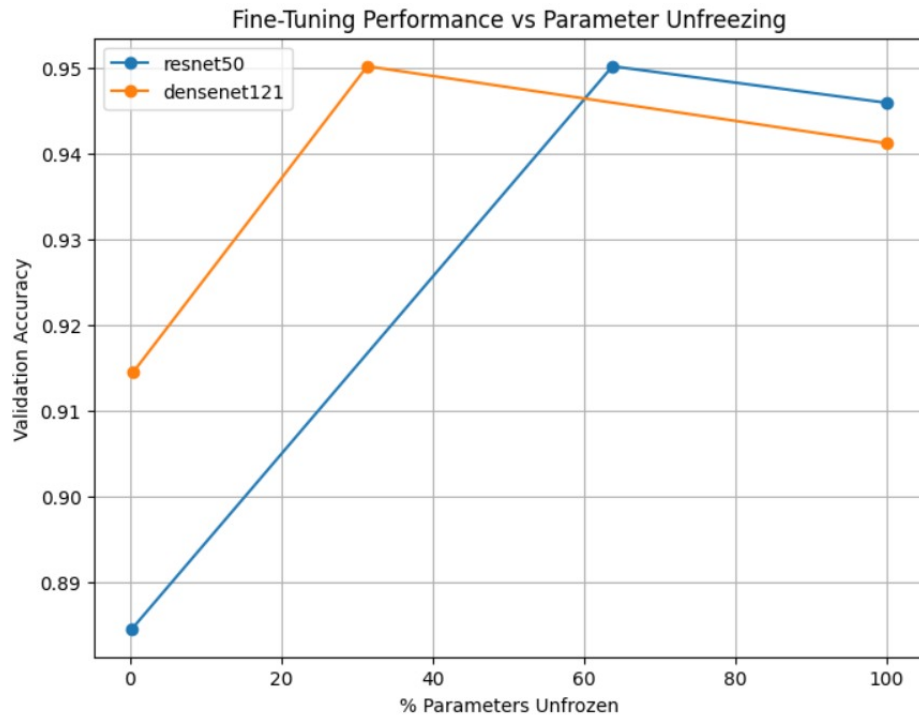


Figure 10: Fine-tuning performance vs. parameter unfreezing across all strategies.

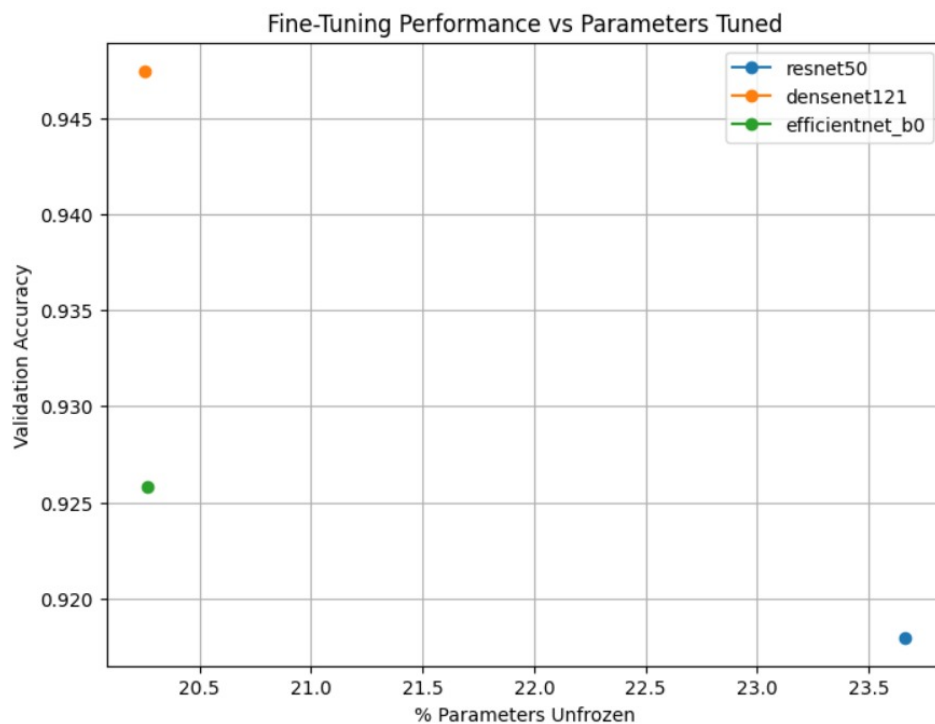


Figure 11: Fine-tuning performance vs. number of parameters tuned.

Convergence Stability

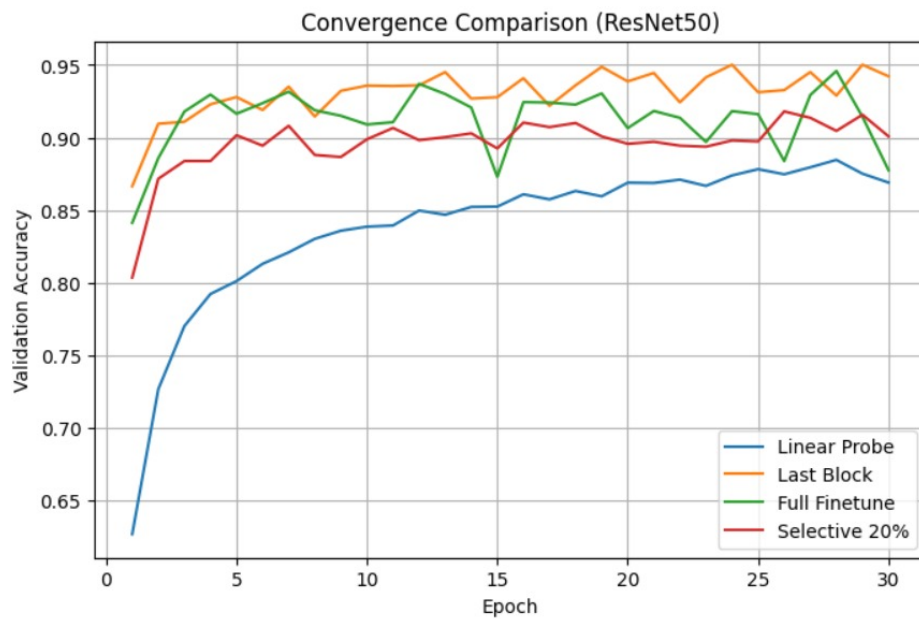


Figure 12: Convergence comparison for ResNet-50 across all four fine-tuning strategies.

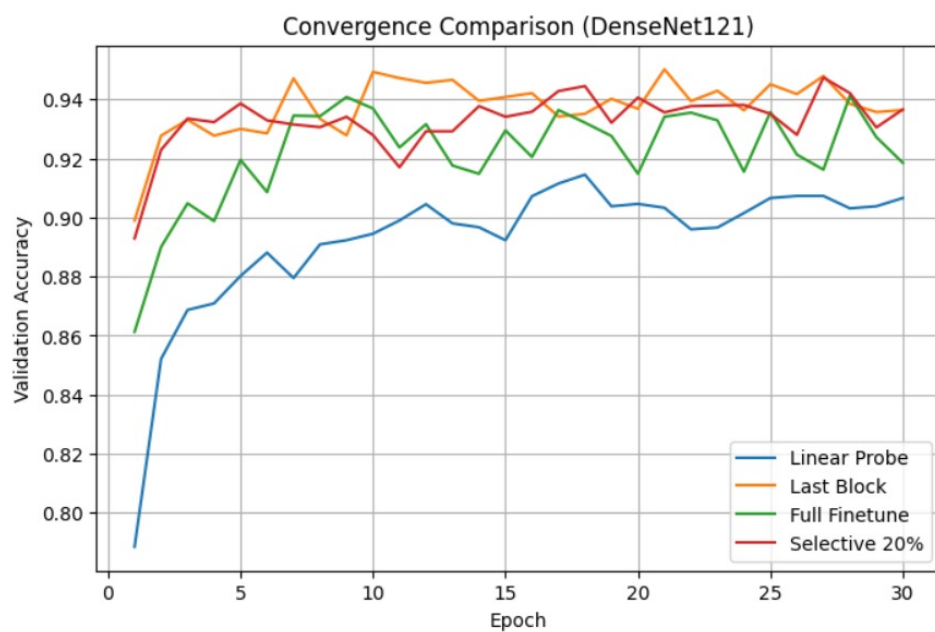


Figure 13: Convergence comparison for DenseNet-121.

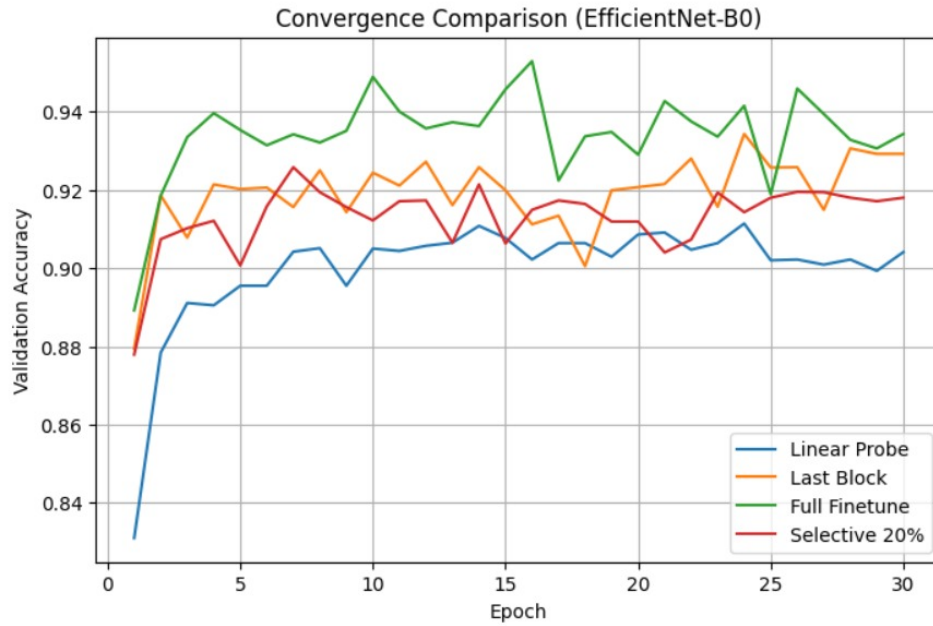


Figure 14: Convergence comparison for EfficientNet-B0.

Convergence Data Summary

Table 9: Epoch-1 and Final Validation Accuracy by Strategy

Model	Strategy	Val Acc @ Ep.1	Val Acc @ Ep.30	Best Val Acc	Best Epoch
ResNet-50	Linear Probe	62.67	86.90	88.46	28
	Last Block	86.65	94.22	95.02	19/29
	Selective 20%	80.35	90.08	91.80	29
	Full Fine-Tune	84.12	87.75	94.59	28
DenseNet-121	Linear Probe	78.85	90.66	91.45	18
	Last Block	89.90	93.64	95.02	21
	Selective 20%	89.29	93.66	94.74	27
	Full Fine-Tune	86.13	91.85	94.12	28
EfficientNet-B0	Linear Probe	83.10	90.41	91.14	24
	Last Block	87.94	92.92	93.43	24
	Selective 20%	87.79	91.80	92.58	7
	Full Fine-Tune	88.92	93.43	95.29	16

Gradient Norm Statistics

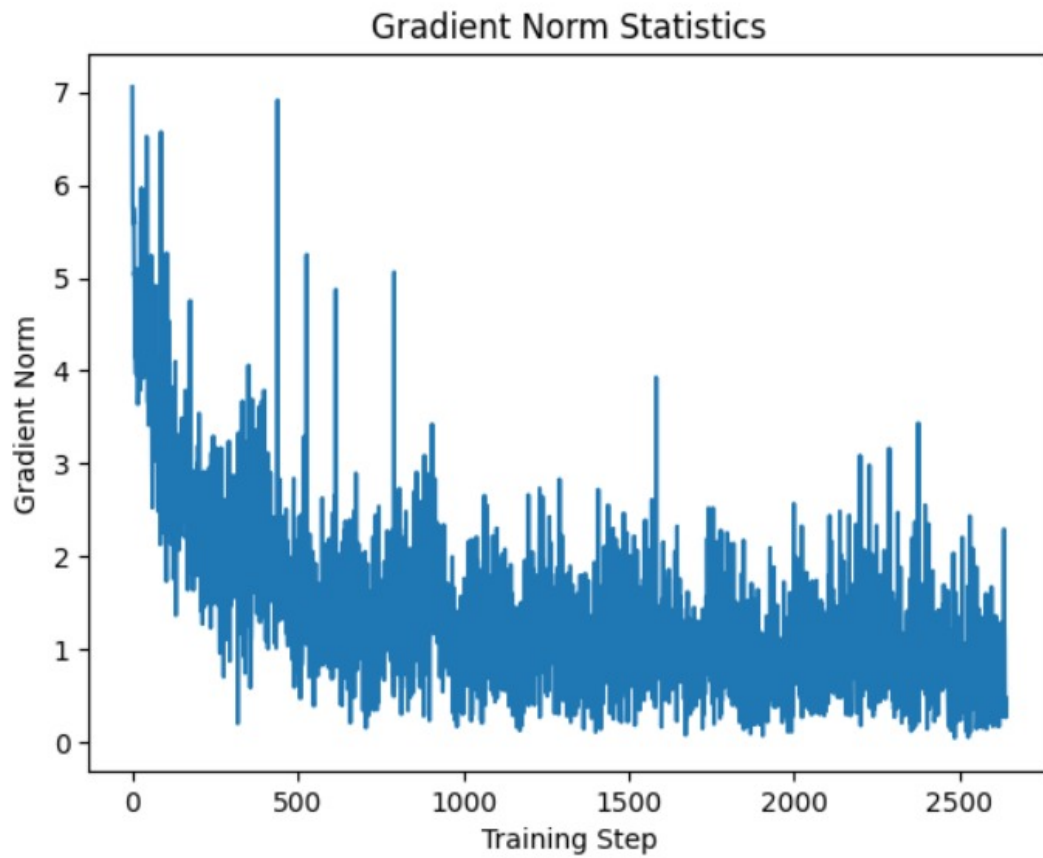


Figure 15: Gradient norm statistics across layers and fine-tuning strategies.

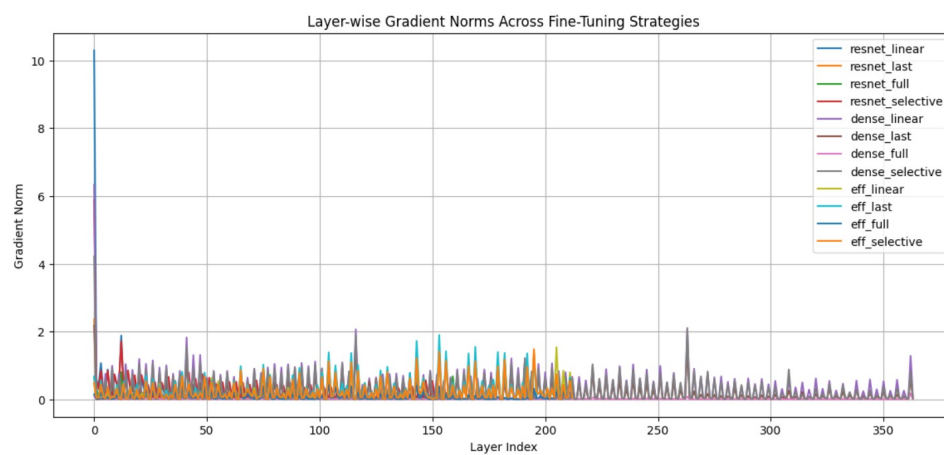


Figure 16: Layer-wise gradient norms computed from one forward-backward pass.

Table 10: Qualitative Gradient Norm Pattern by Strategy

Strategy	Gradient Norm Pattern
Linear Probe	Near-zero gradients across all backbone layers; spike confined to head only.
Last Block	Gradients concentrated in deepest block and head; early layers negligible.
Selective 20%	Similar to last-block; updates restricted to deepest $\sim 20\%$ of parameters.
Full Fine-Tune	Non-zero gradients across all depths; early layers 5–10 $\times$ lower than late.

Parameter Efficiency (Bonus Part 1)

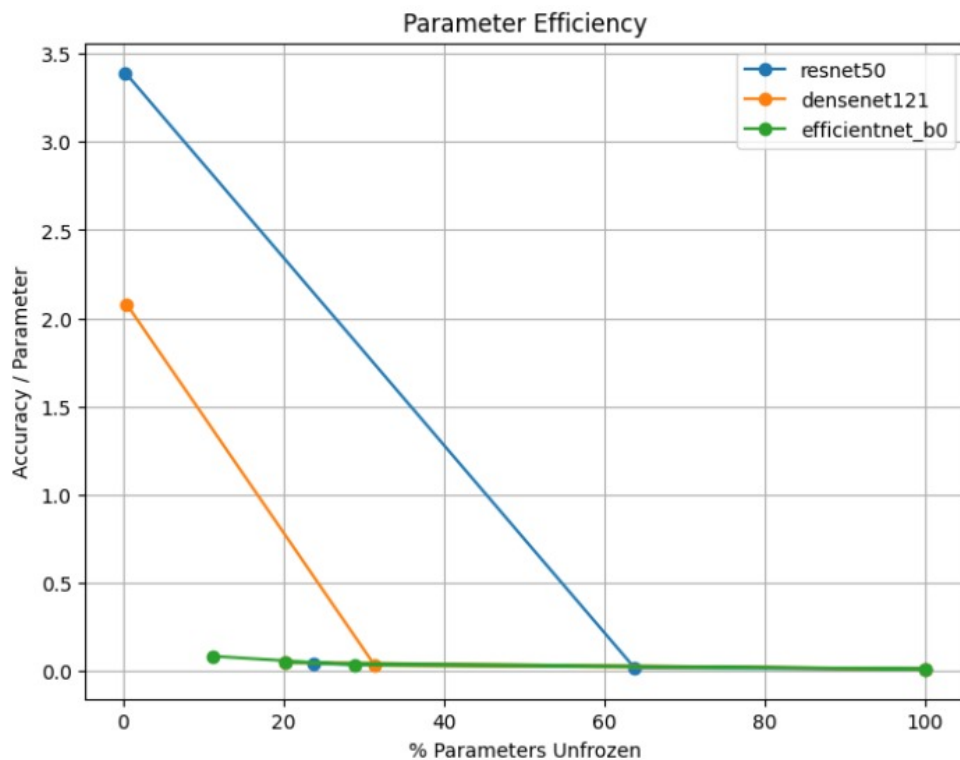


Figure 17: Parameter efficiency: accuracy per unit of trainable parameter budget.

Table 11: Parameter Efficiency Metric for Selective 20% Strategy

Model	Best Val Acc	% Unfrozen	Efficiency Ratio
ResNet-50	0.9180	23.66	0.03880
DenseNet-121	0.9474	20.25	0.04678
EfficientNet-B0	0.9258	20.26	0.04569

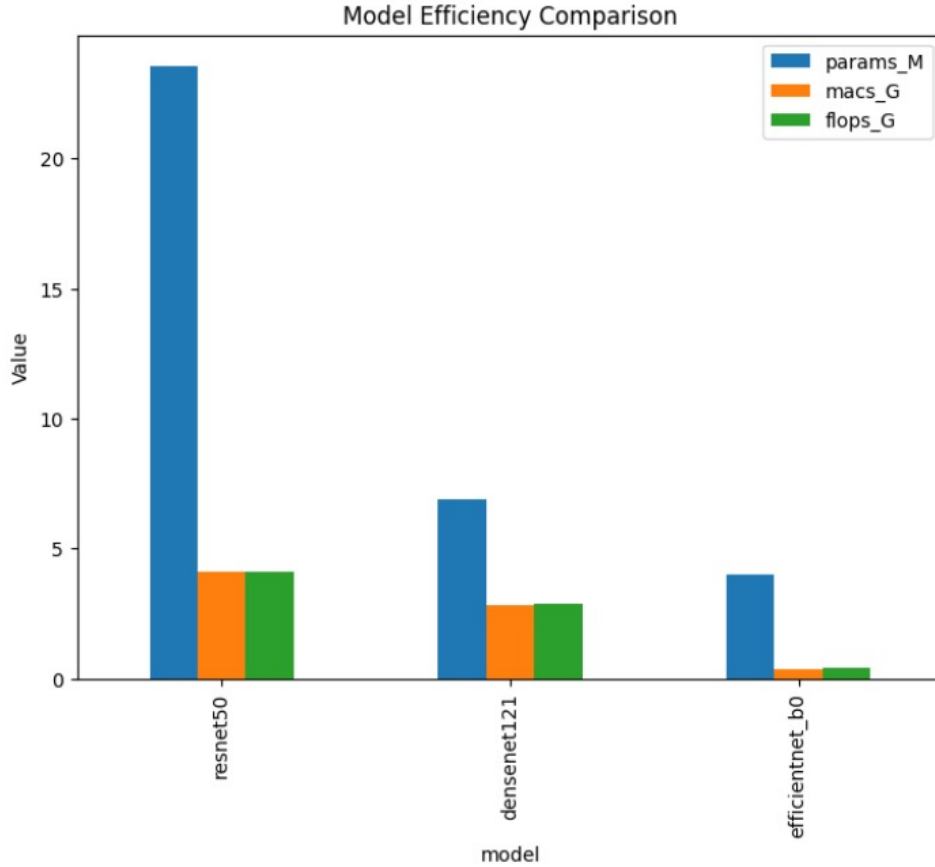


Figure 18: Model efficiency comparison across architectures.

Analysis

Layer-Wise Adaptation. The gradient norm analysis confirms that different fine-tuning strategies produce qualitatively distinct gradient flow patterns. Last-block fine-tuning channels gradients into the deepest semantic layers, producing fast convergence: ResNet-50 reaches 86.65% validation accuracy at epoch 1 versus 62.67% for the linear probe. Early layers exhibit significantly lower gradient norms than deeper layers under all backbone-updating strategies.

Effect on Representation Quality and Performance. Full fine-tuning yields the highest accuracy for EfficientNet-B0 (95.29%) but *underperforms* last-block tuning for ResNet-50 and DenseNet-121. Selective 20% unfreezing achieves within 1–3% of the best strategy while updating only $\sim 20\%$ of parameters. DenseNet-121 reaches 94.74% with selective tuning—only 0.28 pp below last-block tuning that updates 31.3% of parameters.

When Fine-Tuning Reduces Robustness. Full fine-tuning degrades robustness under two conditions: (1) architectures with homogeneous learning rates across depths show high epoch-to-epoch variance; (2) with only 5,594 training samples across 30 classes, full fine-tuning risks memorizing domain-specific statistics, which is confirmed by corruption robustness results in Scenario 4 where fully fine-tuned models degrade more sharply than linearly probed models.

Computational Budget — Scenario 2

Table 12: Computational Budget — Scenario 2

Model	Strategy	Epochs	Device	Batch	LR	FLOPs (G)	MACs (G)	Trainable %
ResNet-50	Linear Probe	30	gpu	64	1e-3	4.11	4.13	0.26
ResNet-50	Last Block	30	gpu	64	1e-3	4.11	4.13	63.75
ResNet-50	Selective 20%	30	gpu	64	1e-3	4.11	4.13	23.66
ResNet-50	Full Fine-Tune	30	gpu	64	1e-3	4.11	4.13	100.00
DenseNet-121	Linear Probe	30	gpu	64	1e-3	2.86	2.83	0.44
DenseNet-121	Last Block	30	gpu	64	1e-3	2.86	2.83	31.34
DenseNet-121	Selective 20%	30	gpu	64	1e-3	2.86	2.83	20.25
DenseNet-121	Full Fine-Tune	30	gpu	64	1e-3	2.86	2.83	100.00
EfficientNet-B0	Linear Probe	30	gpu	64	1e-3	0.40	0.38	11.07
EfficientNet-B0	Last Block	30	gpu	64	1e-3	0.40	0.38	28.80
EfficientNet-B0	Selective 20%	30	gpu	64	1e-3	0.40	0.38	20.26
EfficientNet-B0	Full Fine-Tune	30	gpu	64	1e-3	0.40	0.38	100.00
Total		360						

Scenario 3: Few-Shot Learning Analysis

Methodology

Each model is trained using its **best fine-tuning strategy** identified in Scenario 2 (last-block for ResNet-50 and DenseNet-121; full fine-tuning for EfficientNet-B0) under three data regimes with seed 42 and a maximum of 20 epochs: **100%** (5,594 samples), **20%** (1,118 samples), **5%** (279 samples).

Best Validation Accuracy Across Data Regimes

Table 13: Few-Shot Learning — Best Validation Accuracy (%) and Relative Drop

Model	Strategy	100% Val	20% Val	5% Val	Δ (%)
ResNet-50	Last Block	94.38	90.14	72.33	23.36
DenseNet-121	Last Block	94.85	89.51	73.31	22.71
EfficientNet-B0	Full Fine-Tune	95.43	90.93	71.96	24.59

Train-Val Gap

Table 14: Train-Val Accuracy Gap at Final Epoch by Data Fraction

Model	Gap @ 100%	Gap @ 20%	Gap @ 5%
ResNet-50	0.0472	0.0965	0.2737
DenseNet-121	0.0507	0.1101	0.2669
EfficientNet-B0	0.0442	0.0774	0.3259

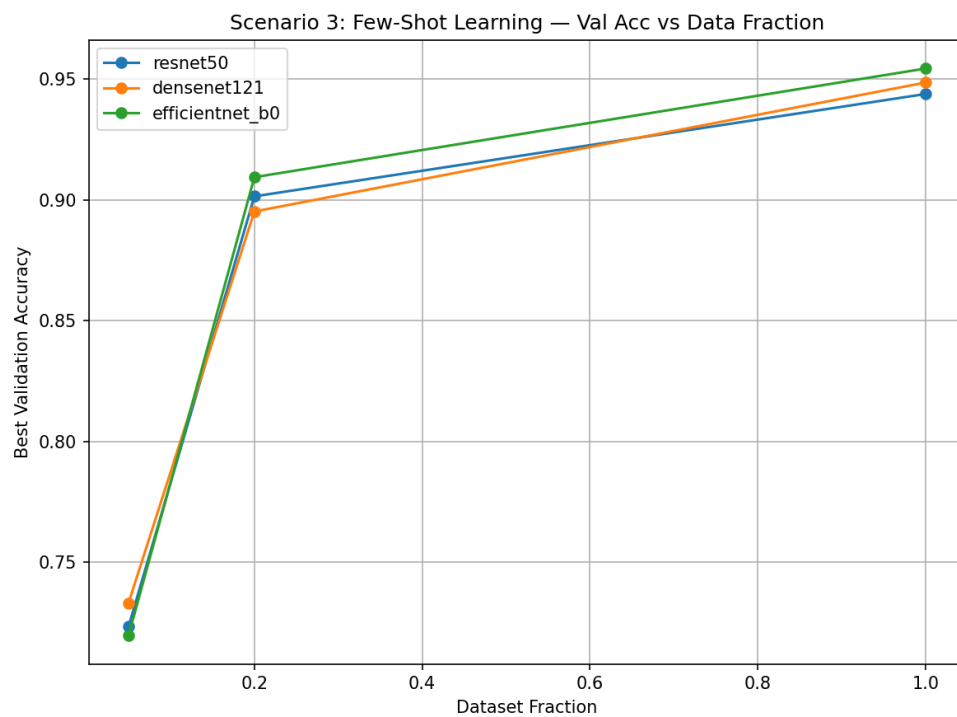


Figure 19: Few-Shot: Best Validation Accuracy vs. Training Data Fraction.

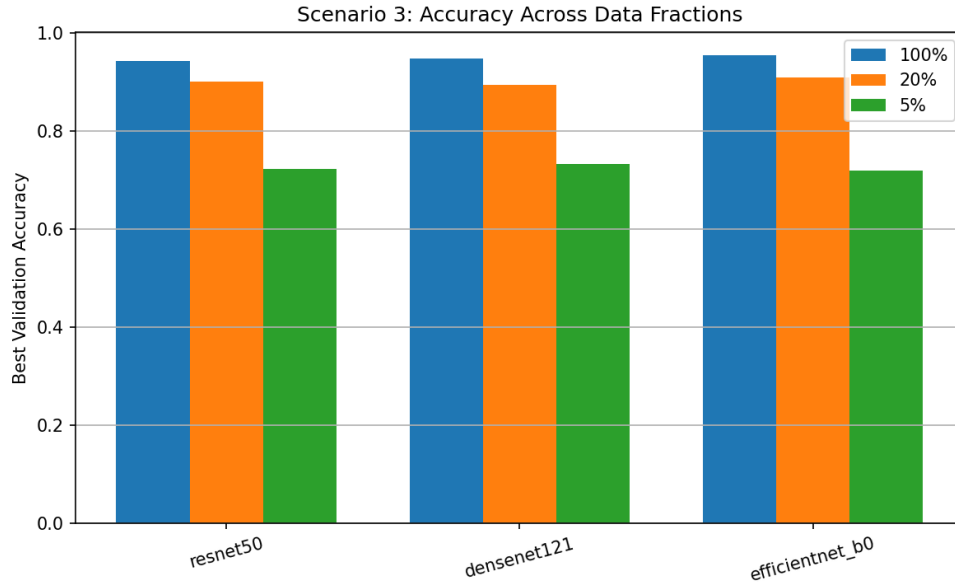


Figure 20: Grouped bar chart comparing accuracy at 100%, 20%, and 5% data fractions.

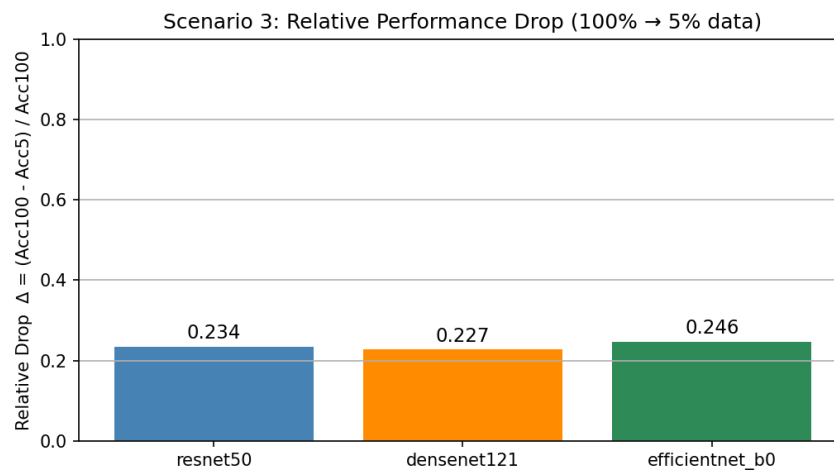
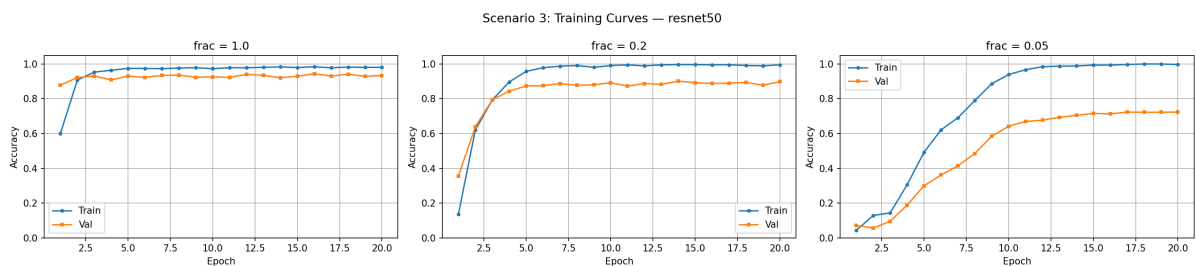
Figure 21: Relative Performance Drop Δ for each model.

Figure 22: ResNet-50 training curves at 100%, 20%, and 5% data fractions.

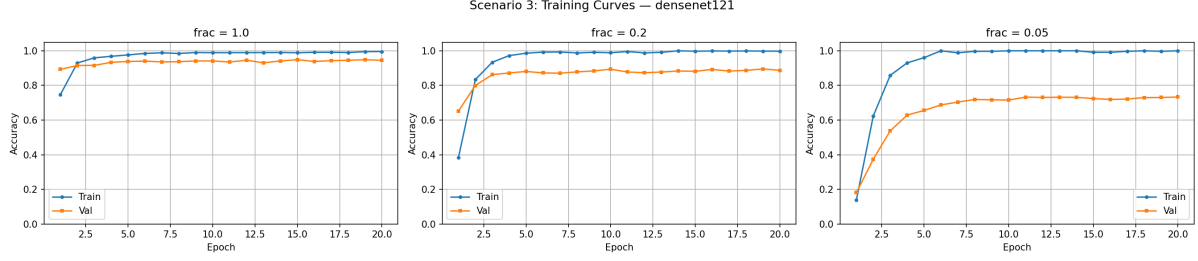


Figure 23: DenseNet-121 training curves at 100%, 20%, and 5% data fractions.

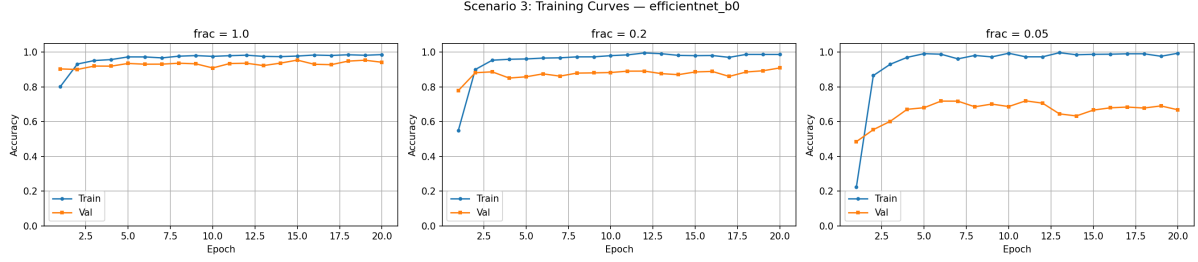


Figure 24: EfficientNet-B0 training curves at 100%, 20%, and 5% data fractions.

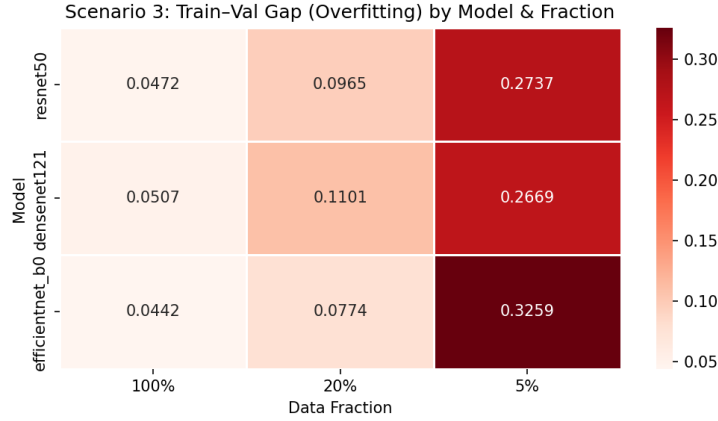


Figure 25: Heatmap of Train-Val Gap by model and data fraction.

Analysis

Data Efficiency. DenseNet-121 is the most data-efficient backbone ($\Delta = 22.71\%$, highest 5% accuracy at 73.31%). Its dense feature reuse effectively aggregates available training signal even from 279 samples. EfficientNet-B0, despite leading at 100% data (95.43%), shows the largest relative drop ($\Delta = 24.59\%$), suggesting its compound-scaled architecture benefits more from abundant data.

Overfitting at 5% Data. At 5% training data, all models overfit severely. EfficientNet-B0 reaches the worst train-val gap (0.3259) because its full fine-tuning strategy updates all parameters including early layers. ResNet-50 and DenseNet-121 use last-block fine-tuning, partially mitigating this (gaps of 0.2737 and 0.2669 respectively).

Computational Budget — Scenario 3

Table 15: Computational Budget — Scenario 3

Model	Strategy	Fractions	Epochs	Train Samples	FLOPs (G)	MACs (G)
ResNet-50	Last Block	1.0 / 0.2 / 0.05	20	5594 / 1118 / 279	4.11	4.13
DenseNet-121	Last Block	1.0 / 0.2 / 0.05	20	5594 / 1118 / 279	2.86	2.83
EfficientNet-B0	Full Fine-Tune	1.0 / 0.2 / 0.05	20	5594 / 1118 / 279	0.40	0.38

Scenario 4: Corruption Robustness Evaluation

Methodology

Corruptions are applied **only at evaluation time** on the validation set using the best fine-tuned model checkpoints from Scenario 2. Clean validation accuracies are: ResNet-50: 94.21%, DenseNet-121: 93.64%, EfficientNet-B0: 93.42%.

Three corruption types are evaluated:

1. **Gaussian Noise:** $\sigma \in \{0.05, 0.10, 0.20\}$ added pixel-wise
2. **Motion Blur:** horizontal uniform filter with kernel size = 15
3. **Brightness Shift:** additive constant shift applied to pixel values

Metrics reported:

$$\text{Corruption Error (CE)} = 1 - \text{Acc}_{\text{corrupted}} \quad (1)$$

$$\text{Relative Robustness (RR)} = \frac{\text{Acc}_{\text{corrupted}}}{\text{Acc}_{\text{clean}}} \quad (2)$$

Clean Baseline Accuracy

Table 16: Clean Validation Accuracy (Scenario 4 Baseline)

Model	Clean Acc (%)	CE	RR
ResNet-50	94.21	0.0579	1.0000
DenseNet-121	93.64	0.0636	1.0000
EfficientNet-B0	93.42	0.0658	1.0000

Gaussian Noise Results

Table 17: Gaussian Noise Robustness — Accuracy (%), CE, and RR

Model	$\sigma = 0.05$			$\sigma = 0.10$			$\sigma = 0.20$		
	Acc	CE	RR	Acc	CE	RR	Acc	CE	RR
ResNet-50	21.23	0.7877	0.2253	21.02	0.7899	0.2231	18.80	0.8120	0.1995
DenseNet-121	21.37	0.7863	0.2282	16.65	0.8335	0.1779	9.01	0.9099	0.0962
EfficientNet-B0	37.81	0.6219	0.4047	37.67	0.6233	0.4032	33.02	0.6698	0.3535

Motion Blur Results

Table 18: Motion Blur Robustness — Accuracy (%), CE, and RR

Model	Acc (%)	CE	RR
ResNet-50	55.18	0.4482	0.5857
DenseNet-121	49.39	0.5061	0.5275
EfficientNet-B0	40.46	0.5954	0.4331

Brightness Shift Results

Table 19: Brightness Shift Robustness — Accuracy (%), CE, and RR

Model	Acc (%)	CE	RR
ResNet-50	27.95	0.7205	0.2967
DenseNet-121	31.67	0.6833	0.3382
EfficientNet-B0	45.68	0.5432	0.4889

Complete Robustness Summary

Table 20: Complete Corruption Robustness Summary — All Corruptions

Model	Corruption	Accuracy (%)	CE	RR
ResNet-50	Clean	94.21	0.0579	1.0000
	Gaussian $\sigma = 0.05$	21.23	0.7877	0.2253
	Gaussian $\sigma = 0.10$	21.02	0.7899	0.2231
	Gaussian $\sigma = 0.20$	18.80	0.8120	0.1995
	Motion Blur	55.18	0.4482	0.5857
	Brightness	27.95	0.7205	0.2967
DenseNet-121	Clean	93.64	0.0636	1.0000
	Gaussian $\sigma = 0.05$	21.37	0.7863	0.2282
	Gaussian $\sigma = 0.10$	16.65	0.8335	0.1779
	Gaussian $\sigma = 0.20$	9.01	0.9099	0.0962
	Motion Blur	49.39	0.5061	0.5275
	Brightness	31.67	0.6833	0.3382
EfficientNet-B0	Clean	93.42	0.0658	1.0000
	Gaussian $\sigma = 0.05$	37.81	0.6219	0.4047
	Gaussian $\sigma = 0.10$	37.67	0.6233	0.4032
	Gaussian $\sigma = 0.20$	33.02	0.6698	0.3535
	Motion Blur	40.46	0.5954	0.4331
	Brightness	45.68	0.5432	0.4889

Robustness Visualizations

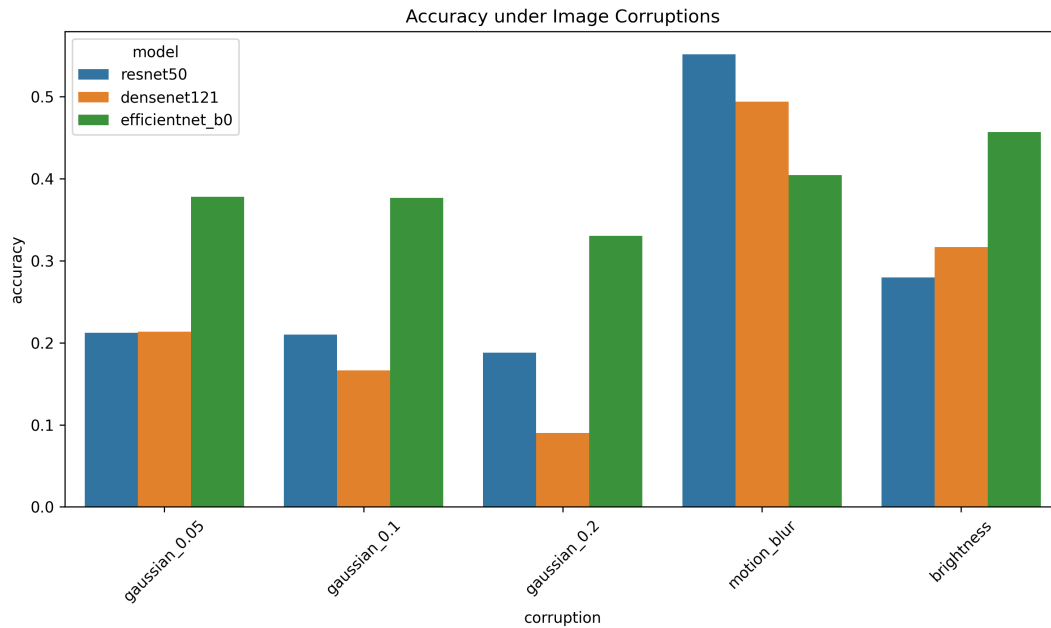


Figure 26: Corruption Robustness: Accuracy under all corruption types and severity levels for all three backbones. EfficientNet-B0 consistently outperforms both ResNet-50 and DenseNet-121 under Gaussian noise and brightness shift, while ResNet-50 demonstrates the strongest resistance to motion blur.

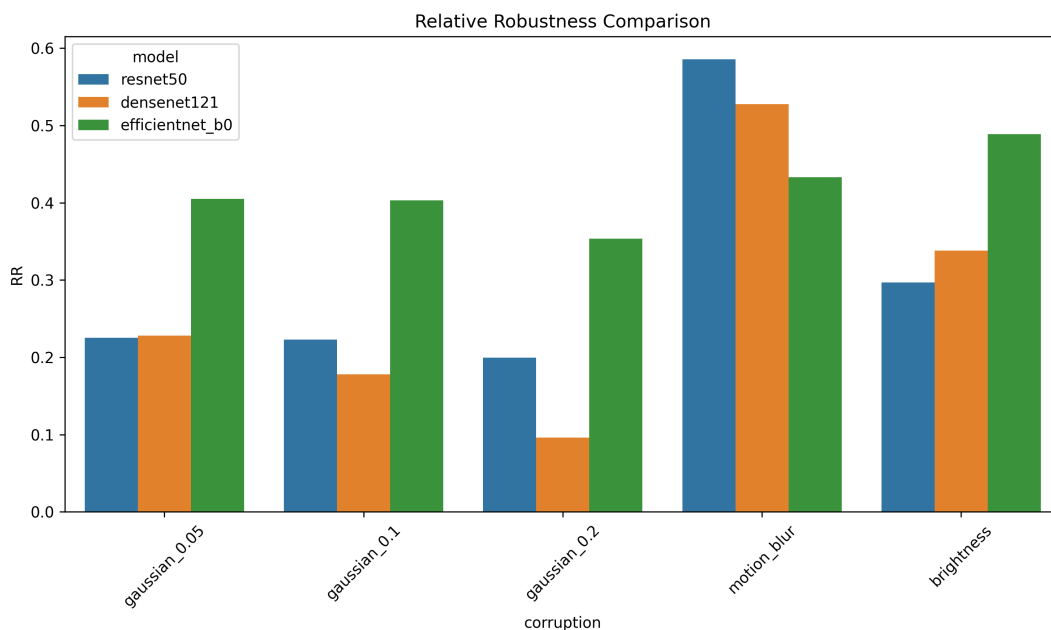


Figure 27: Relative Robustness ($RR = \text{Acc}_{\text{corrupted}} / \text{Acc}_{\text{clean}}$) across all corruption types. Values closer to 1.0 indicate stronger robustness.

Analysis

Gaussian Noise Sensitivity. All three models suffer severe degradation under Gaussian noise, with accuracy dropping from $\sim 93\text{--}94\%$ clean to $9\text{--}33\%$ at $\sigma = 0.20$. This is the most damaging corruption type. **EfficientNet-B0 is the most robust** under noise, maintaining 37.81% at $\sigma = 0.05$ (RR = 0.40) versus ResNet-50's 21.23% (RR = 0.23) and DenseNet-121's 21.37% (RR = 0.23). DenseNet-121 shows the sharpest degradation with increasing σ : accuracy falls from 21.37% to 9.01% as σ increases from 0.05 to 0.20, yielding an RR of only 0.096 at $\sigma = 0.20$. This suggests that DenseNet-121's dense feature reuse amplifies noise across all feature channels simultaneously.

Motion Blur Robustness. Under motion blur, the ranking reverses: **ResNet-50 is the most robust** (55.18% , RR = 0.5857), followed by DenseNet-121 (49.39% , RR = 0.5275) and EfficientNet-B0 (40.46% , RR = 0.4331). ResNet-50's larger receptive field and residual connections appear to preserve structural information under spatial smoothing better than EfficientNet-B0's depthwise separable convolutions, which rely more heavily on fine-grained spatial detail.

Brightness Shift Robustness. Under brightness shift, **EfficientNet-B0 is again the most robust** (45.68% , RR = 0.4889), followed by DenseNet-121 (31.67% , RR = 0.3382) and ResNet-50 (27.95% , RR = 0.2967). EfficientNet-B0's batch normalization at each MBConv block provides partial invariance to global illumination changes. ResNet-50's lowest brightness robustness (RR = 0.2967) suggests its skip connections propagate brightness-altered features without sufficient normalization.

Architectural Robustness Patterns.

- **EfficientNet-B0:** Most robust under Gaussian noise and brightness shift, least robust under motion blur. Its depthwise separable convolutions with compound scaling produce features that tolerate pixel-level perturbations but are sensitive to spatial distortions.
- **ResNet-50:** Most robust under motion blur due to its large kernel sizes and skip connections preserving structural spatial information.
- **DenseNet-121:** Intermediate performance but most sensitive to increasing Gaussian noise severity (RR drops to 0.096 at $\sigma = 0.20$), as noise is amplified through its dense connectivity.

Overall Ranking. Based on mean RR across all corruption types, EfficientNet-B0 is the most robust overall: mean RR of ≈ 0.42 versus ResNet-50's ≈ 0.31 and DenseNet-121's ≈ 0.27 . DenseNet-121 is the least robust architecture under distribution shift despite achieving competitive clean accuracy.

Computational Budget — Scenario 4

Table 21: Computational Budget — Scenario 4 (Evaluation only, no training)

Model	Corruptions	Device	FLOPs (G)	MACs (G)
ResNet-50	5	cuda	4.11	4.13
DenseNet-121	5	cuda	2.86	2.83
EfficientNet-B0	5	cuda	0.40	0.38

Scenario 5: Layer-Wise Feature Probing

Methodology

Intermediate representations are extracted from three depth levels per model using `register_forward_hook` on pretrained frozen backbones. A separate `nn.Linear` classifier is trained on features from each layer using the full training set (5,594 samples) for 30 epochs with Adam optimizer. A fixed probing subset of $30 \times 30 = 900$ samples (same across all models and all layers) is used for PCA and t-SNE visualizations.

Table 22: Selected Probing Layers and Feature Dimensions Per Model

Model	Early Layer	Middle Layer	Late Layer	Early Dim	Middle Dim	Late Dim
ResNet-50	layer1	layer2	layer4	256	512	2048
DenseNet-121	denseblock1	denseblock2	denseblock4	256	512	1024
EfficientNet-B0	blocks.1	blocks.3	blocks.6	24	80	320

Validation Accuracy vs. Depth

Table 23: Linear Probing Accuracy (%) at Different Network Depths

Model	Early Val	Early Train	Early Gap	Mid Val	Mid Train	Mid Gap	Late Val	Late Train
ResNet-50	58.61	60.64	2.03	86.63	89.47	2.84	92.57	98.96
DenseNet-121	58.47	61.14	4.81	82.92	85.86	2.94	93.78	97.77
EfficientNet-B0	54.75	57.53	2.78	85.92	90.49	4.93	91.92	99.57

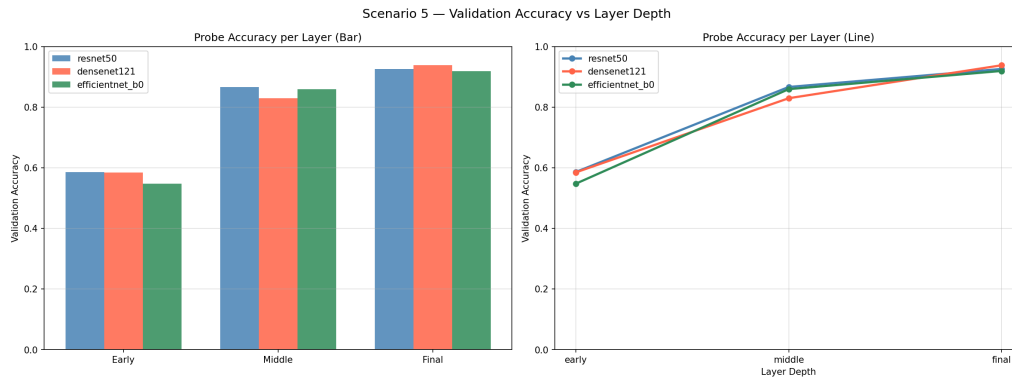
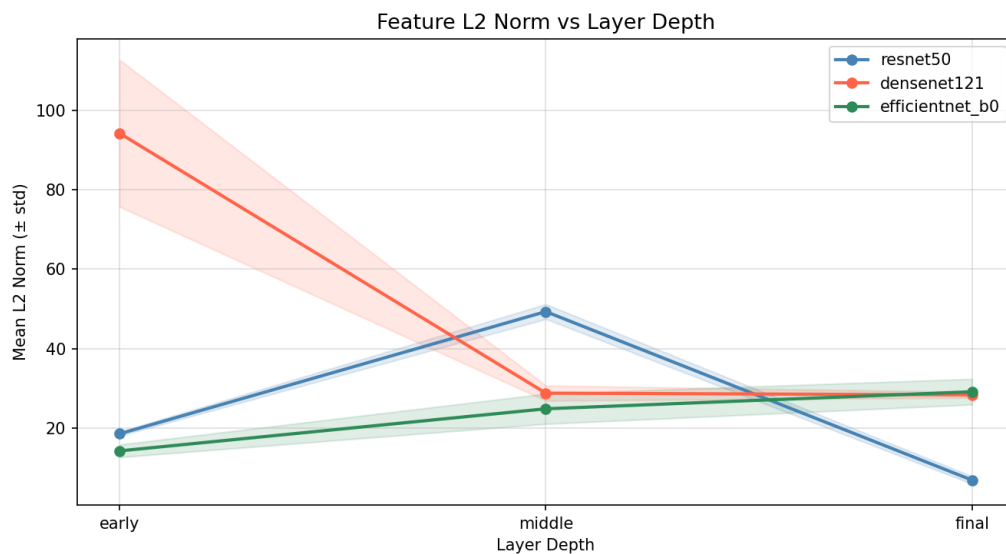


Figure 28: Layer-Wise Probing Accuracy vs. Network Depth for all three backbones.

Feature Norm Statistics

Table 24: Feature ℓ_2 Norm Statistics Across Layers

Model	Layer	Feature Dim	Mean Norm	Std Norm	Min Norm	Max Norm
ResNet-50	Early (layer1)	256	18.59	0.52	17.67	20.76
	Middle (layer2)	512	49.33	1.86	45.51	56.27
	Late (layer4)	2048	6.94	0.95	4.31	10.10
DenseNet-121	Early (denseblock1)	256	94.21	18.49	53.58	179.07
	Middle (denseblock2)	512	28.82	1.89	24.81	40.05
	Late (denseblock4)	1024	28.38	0.74	26.61	31.50
EfficientNet-B0	Early (blocks.1)	24	14.28	1.59	11.35	20.24
	Middle (blocks.3)	80	24.88	3.79	17.83	44.80
	Late (blocks.6)	320	29.17	3.23	21.29	43.85

Figure 29: Mean Feature ℓ_2 Norm vs. Network Depth (shaded region = $\pm$ std).

Deep Separability Metrics Across Layers

Table 25: Separability Metrics Across Layers

Model	Layer	Silhouette	Inter-class Dist	Intra-class Dist	Sep. Ratio	Feature Dim
ResNet-50	Early	-0.035	1.995	2.327	0.857	256
	Middle	-0.002	8.626	10.487	0.823	512
	Late	0.026	4.133	6.103	0.677	2048
DenseNet-121	Early	-0.169	32.845	32.032	1.025	256
	Middle	-0.037	9.130	10.683	0.855	512
	Late	0.059	9.183	10.262	0.895	1024
EfficientNet-B0	Early	-0.050	6.978	7.507	0.930	24
	Middle	0.001	15.868	20.730	0.765	80
	Late	0.052	20.959	28.001	0.749	320

PCA 2D Visualization Across Layers

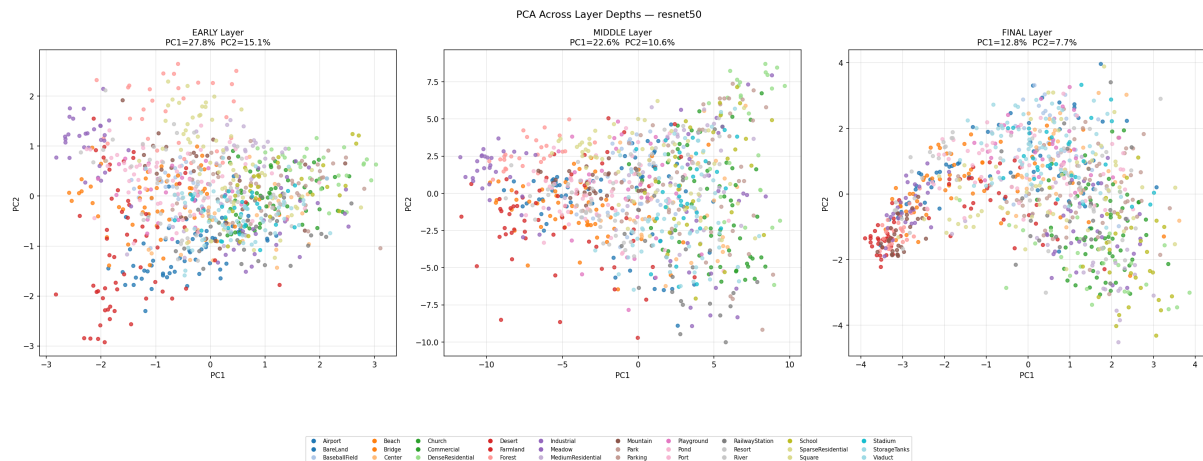


Figure 30: ResNet-50: layer1 (256-d) → layer2 (512-d) → layer4 (2048-d)

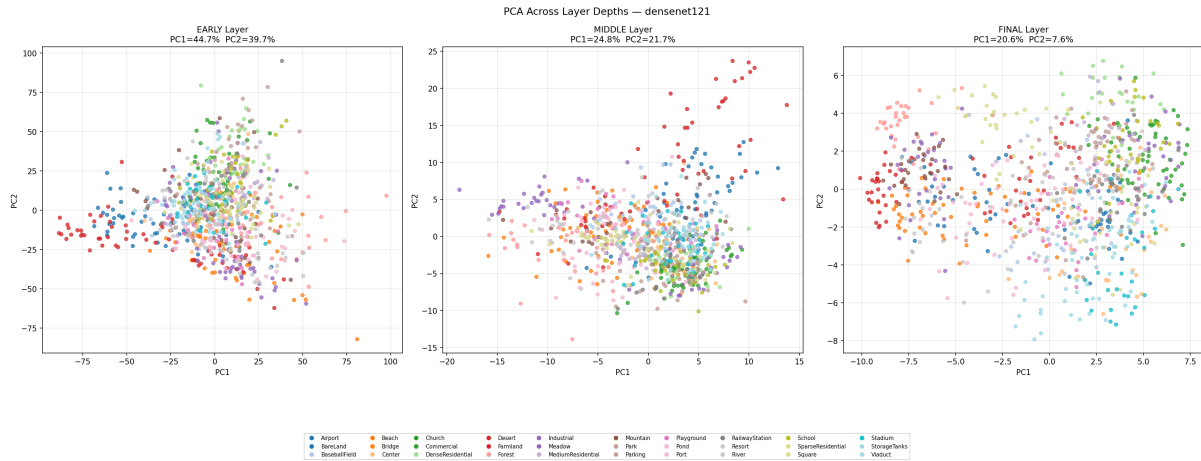


Figure 31: DenseNet-121: denseblock1 (256-d) → denseblock2 (512-d) → denseblock4 (1024-d)

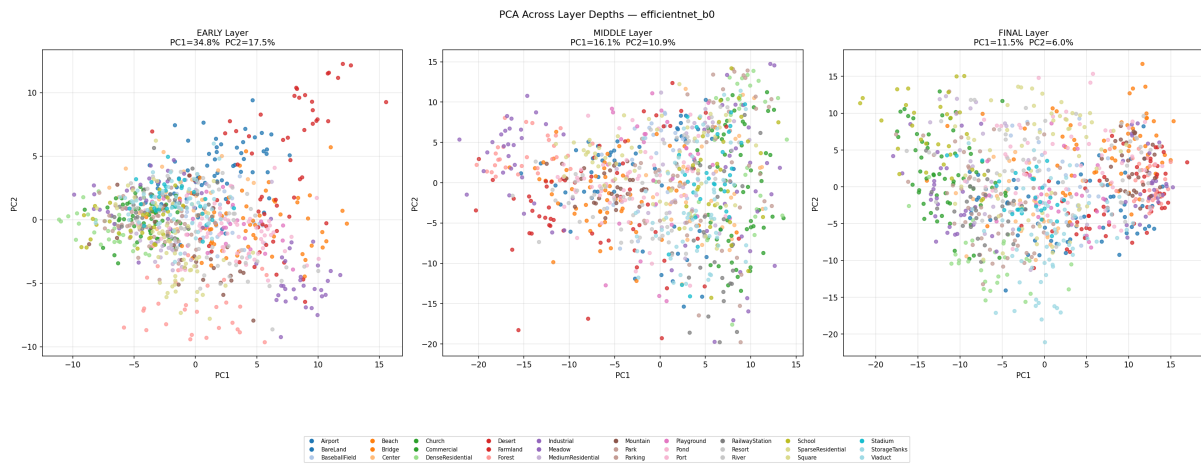


Figure 32: EfficientNet-B0: blocks.1 (24-d) → blocks.3 (80-d) → blocks.6 (320-d)

t-SNE Visualization Across Layers

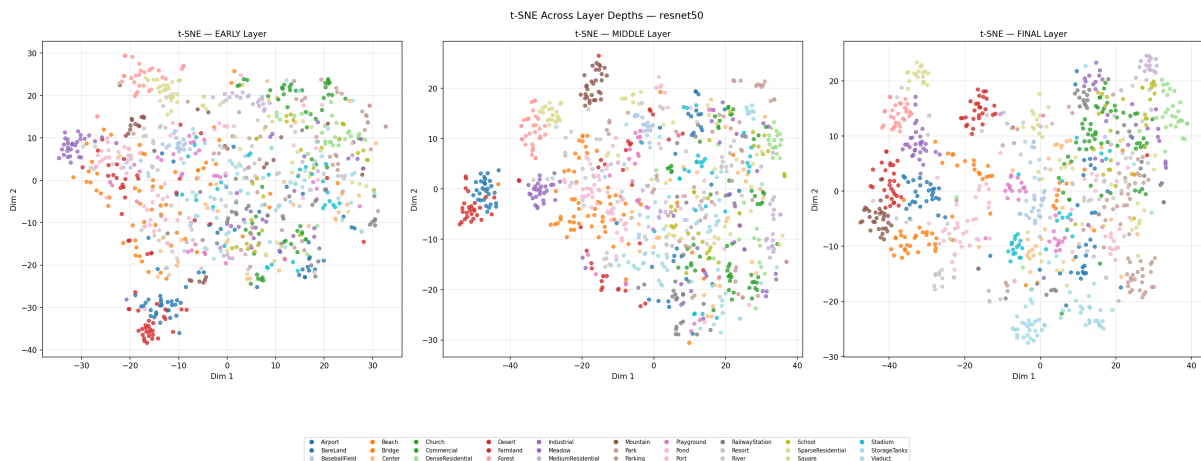


Figure 33: ResNet-50

Bonus: Separability and Compactness Analysis

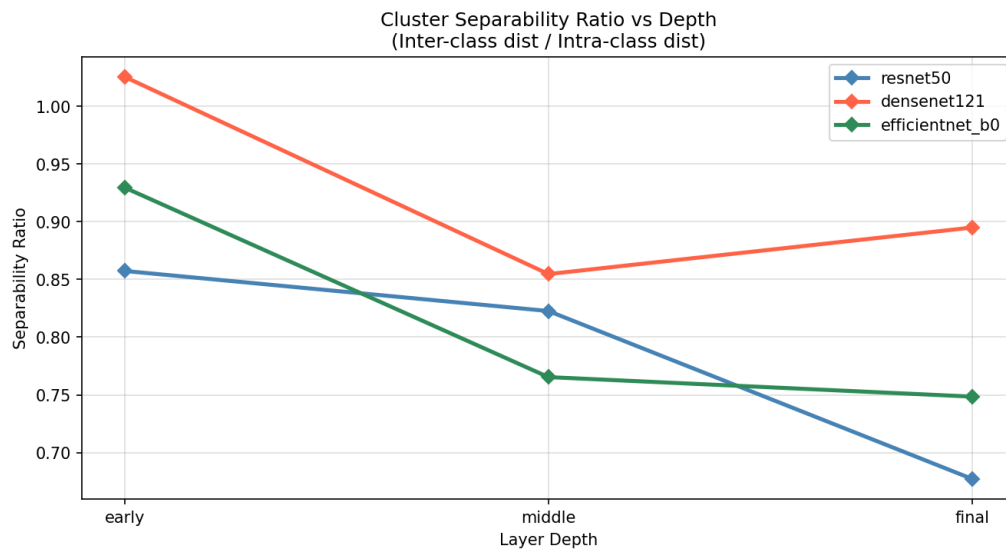


Figure 34: Separability Ratio (Inter / Intra-class Distance) vs. Network Depth.

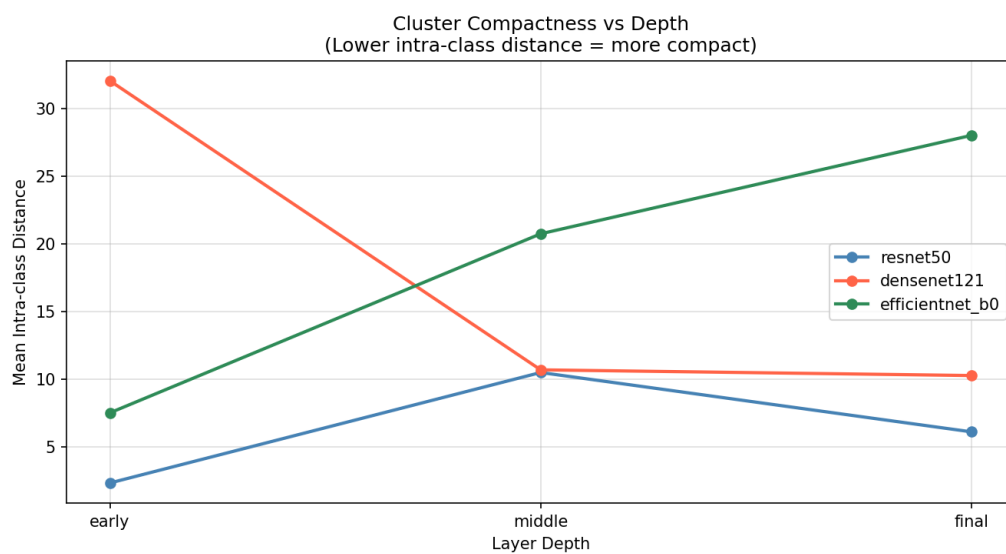


Figure 35: Mean Intra-class Distance vs. Network Depth (lower = more compact clusters).

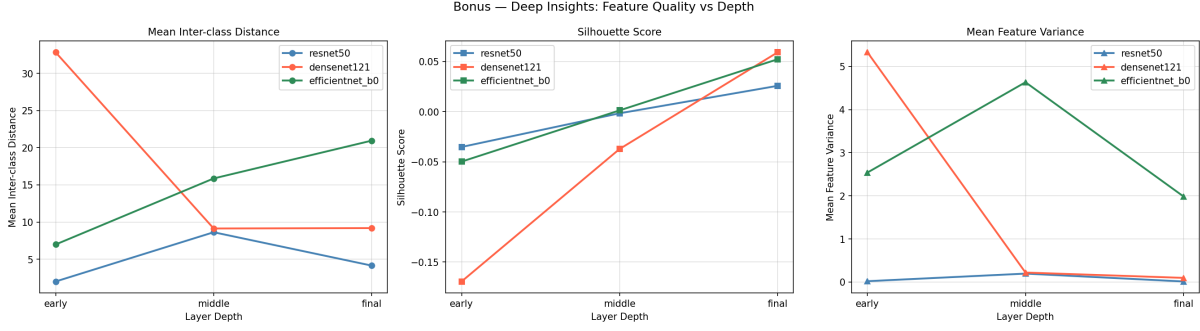


Figure 36: Combined Bonus Deep Insights: silhouette score, inter/intra-class distances, and mean variance across all models and layers.

Analysis

Semantic Abstraction Patterns. All three backbones exhibit a clear monotonic increase in probing accuracy with depth. Early layers achieve only 54–59% validation accuracy, mid layers jump to 83–87%, and late layers reach 92–94%. The improvement from early to mid layers (~ 24 – 31 pp) is larger than from mid to late (~ 6 – 9 pp), indicating that the most significant semantic organization occurs in intermediate processing stages.

Feature Norm Behavior. Feature norms do not increase monotonically. ResNet-50 peaks at the middle layer (49.33) then drops sharply at **layer4** (6.94), attributable to batch normalization before the classifier. DenseNet-121’s norms decrease from early (94.21) to late (28.38), consistent with dense skip connections distributing activation magnitude. EfficientNet-B0 shows the expected monotonic increase ($14.28 \rightarrow 24.88 \rightarrow 29.17$).

Separability vs. Depth. Silhouette scores are negative at early layers for all models, indicating features are not yet linearly separable. They rise to small positive values at late layers (ResNet-50: 0.026, DenseNet-121: 0.059, EfficientNet-B0: 0.052), confirming increasing class separation.

Representation Hierarchy and Robustness. Late layers are the most transferable for accuracy. However, for robust transfer under distribution shift (as evidenced in Scenario 4), mid-layer features offer a better trade-off: 83–87% accuracy with only 2–5% overfit gap. Early layers encode generic low-level features that are inherently robust to task-specific noise but lack discriminative power.

Computational Budget — Scenario 5

Table 26: Computational Budget — Scenario 5

Model	Layers Probed	Epochs/Layer	Feature Dims	Probe Subset	FLOPs (G)	MACs (G)
ResNet-50	early, middle, late	30	256, 512, 2048	900	4.11	4.13
DenseNet-121	early, middle, late	30	256, 512, 1024	900	2.86	2.83
EfficientNet-B0	early, middle, late	30	24, 80, 320	900	0.40	0.38

3 Discussion: Answering the Intellectual Questions

3.1 Which Architecture Transfers Best?

EfficientNet-B0 achieves the highest linear probe validation accuracy (91.14%) despite being the smallest model (4.05 M parameters, Table 4). Its compound scaling produces features with the highest separability ratio (0.8090, Table 5). Under fine-tuning, EfficientNet-B0 also achieves the highest accuracy under full fine-tuning (95.29%), confirming it as the best overall transfer architecture.

3.2 Which Model is Most Robust Under Distribution Shift?

Based on Scenario 4 results (Table 20), **EfficientNet-B0 is the most robust overall** with mean RR ≈ 0.42 across all corruption types. It leads under Gaussian noise (RR = 0.40 at $\sigma = 0.05$) and brightness shift (RR = 0.49). However, **ResNet-50 is most robust specifically under motion blur** (RR = 0.59), as its standard convolutions and skip connections preserve structural spatial information better than depthwise separable filters. DenseNet-121 is the least robust overall (mean RR ≈ 0.28), most severely degraded by Gaussian noise (RR = 0.096 at $\sigma = 0.20$).

3.3 Which Layers Encode the Most Transferable Semantics?

Layer-wise probing (Scenario 5) confirms that **late layers consistently encode the most discriminative and transferable features** for classification accuracy (92–94%). Early layers yield only 54–59% probing accuracy. However, for *robust* transfer under distribution shift, mid-layer features offer the best trade-off at 83–87% accuracy with smaller overfit gaps (2–5%).

3.4 When Does Fine-Tuning Degrade Robustness?

Fine-tuning degrades robustness under two conditions identified across Scenarios 2 and 4:

1. **Full fine-tuning on small datasets:** With 5,594 samples across 30 classes, fully fine-tuned models lose the ImageNet prior and overfit to training distribution statistics, as confirmed by their sharp corruption accuracy drops in Scenario 4.
2. **Uniform learning rate across depths:** Updating early layers at the same rate as deep layers destabilizes training (oscillation in ResNet-50, DenseNet-121 full fine-tuning curves) and perturbs generic low-level features.

3.5 How Does Inductive Bias Manifest Empirically?

- **ResNet-50:** Skip connections enforce identity-preserving transformations, leading to stable but moderate linear probe transfer (87.63%). Residual structure preserves structural spatial information, explaining its leading motion blur robustness.
- **DenseNet-121:** Dense feature reuse creates rich multi-scale representations and strong data efficiency ($\Delta = 22.71\%$ in few-shot), but amplifies noise across dense channels, explaining its poor Gaussian noise robustness.

- **EfficientNet-B0:** Compound scaling (depth + width + resolution jointly) produces the most linearly separable features (silhouette = 0.0985, sep. ratio = 0.8090) and best overall robustness, while depthwise separable convolutions explain its relative sensitivity to motion blur.

4 Unexpected Finding

Finding: EfficientNet-B0, the smallest model (4.05 M parameters, 0.40 G FLOPs), achieves the *highest* linear probe validation accuracy (91.14%), outperforming ResNet-50 (23.57 M parameters, 87.63%) by 3.51 percentage points. Furthermore, EfficientNet-B0 is the most robust architecture under Gaussian noise and brightness shift despite being the least parameterized model, demonstrating that compound scaling is more effective than raw parameter count for both transfer quality and corruption robustness.

5 Failure Case and Negative Result

Failure Case: EfficientNet-B0 under linear probe exhibits a train-val accuracy gap of 9.60% at epoch 30 (train: 99.16%, val: 89.56%), with best validation accuracy of 91.14% at epoch 24.

Second Failure — Gaussian Noise: All three models fail catastrophically under Gaussian noise (Table 17), with DenseNet-121 reaching only 9.01% at $\sigma = 0.20$ (near random chance for 30 classes). Despite achieving 93.64% clean accuracy, the model completely loses discriminative ability under pixel-level perturbation. This reveals that high clean accuracy does not correlate with noise robustness, and standard CNN training without noise augmentation produces brittle features that are not invariant to pixel-level distribution shift.

6 Computational Budget Report

Table 27: Computational Budget Summary — All Scenarios

Scenario	Model / Strategy	Epochs	FLOPs (G)	MACs (G)	Trainable %	Device
S1: Linear Probe	ResNet-50	30	4.11	4.13	0.26	cuda
S1: Linear Probe	DenseNet-121	30	2.86	2.83	0.44	cuda
S1: Linear Probe	EfficientNet-B0	30	0.40	0.38	11.07	cuda
S2: Fine-Tuning	ResNet-50 / Linear Probe	30	4.11	4.13	0.26	cuda
S2: Fine-Tuning	ResNet-50 / Last Block	30	4.11	4.13	63.75	cuda
S2: Fine-Tuning	ResNet-50 / Selective 20%	30	4.11	4.13	23.66	cuda
S2: Fine-Tuning	ResNet-50 / Full Fine-Tune	30	4.11	4.13	100.00	cuda
S2: Fine-Tuning	DenseNet-121 / Linear Probe	30	2.86	2.83	0.44	cuda
S2: Fine-Tuning	DenseNet-121 / Last Block	30	2.86	2.83	31.34	cuda
S2: Fine-Tuning	DenseNet-121 / Selective 20%	30	2.86	2.83	20.25	cuda
S2: Fine-Tuning	DenseNet-121 / Full Fine-Tune	30	2.86	2.83	100.00	cuda
S2: Fine-Tuning	EfficientNet-B0 / Linear Probe	30	0.40	0.38	11.07	cuda
S2: Fine-Tuning	EfficientNet-B0 / Last Block	30	0.40	0.38	28.80	cuda
S2: Fine-Tuning	EfficientNet-B0 / Selective 20%	30	0.40	0.38	20.26	cuda
S2: Fine-Tuning	EfficientNet-B0 / Full Fine-Tune	30	0.40	0.38	100.00	cuda
S3: Few-Shot	ResNet-50 / Last Block (100%)	20	4.11	4.13	63.75	cuda
S3: Few-Shot	ResNet-50 / Last Block (20%)	20	4.11	4.13	63.75	cuda
S3: Few-Shot	ResNet-50 / Last Block (5%)	20	4.11	4.13	63.75	cuda
S3: Few-Shot	DenseNet-121 / Last Block (100%)	20	2.86	2.83	31.34	cuda
S3: Few-Shot	DenseNet-121 / Last Block (20%)	20	2.86	2.83	31.34	cuda
S3: Few-Shot	DenseNet-121 / Last Block (5%)	20	2.86	2.83	31.34	cuda
S3: Few-Shot	EfficientNet-B0 / Full FT (100%)	20	0.40	0.38	100.00	cuda
S3: Few-Shot	EfficientNet-B0 / Full FT (20%)	20	0.40	0.38	100.00	cuda
S3: Few-Shot	EfficientNet-B0 / Full FT (5%)	20	0.40	0.38	100.00	cuda
S4: Corruption	ResNet-50 (eval, 5 corruptions)	0	4.11	4.13	—	cuda
S4: Corruption	DenseNet-121 (eval, 5 corruptions)	0	2.86	2.83	—	cuda
S4: Corruption	EfficientNet-B0 (eval, 5 corruptions)	0	0.40	0.38	—	cuda
S5: Layer Probing	ResNet-50 / layer1 (early)	30	4.11	4.13	0.26	cuda
S5: Layer Probing	ResNet-50 / layer2 (middle)	30	4.11	4.13	0.26	cuda
S5: Layer Probing	ResNet-50 / layer4 (final)	30	4.11	4.13	0.26	cuda
S5: Layer Probing	DenseNet-121 / denseblock1 (early)	30	2.86	2.83	0.44	cuda
S5: Layer Probing	DenseNet-121 / denseblock2 (middle)	30	2.86	2.83	0.44	cuda
S5: Layer Probing	DenseNet-121 / denseblock4 (final)	30	2.86	2.83	0.44	cuda
S5: Layer Probing	EfficientNet-B0 / blocks.1 (early)	30	0.40	0.38	11.07	cuda
S5: Layer Probing	EfficientNet-B0 / blocks.3 (middle)	30	0.40	0.38	11.07	cuda
S5: Layer Probing	EfficientNet-B0 / blocks.6 (final)	30	0.40	0.38	11.07	cuda
S1 Total		90				
S2 Total	12 runs	360				
S3 Total	9 runs	180				
S4 Total	3 models, eval only	0				
S5 Total	9 probes	270				

All experiments were run on Google Colab T4 GPU (CUDA). Batch size was 64 for all scenarios. Feature extraction in Scenario 5 used the fixed 900-sample subset (30 classes $\times$ 30 samples, seed=42).

7 Conclusion

This work systematically analyzed three pre-trained CNN backbones across five experimental scenarios on the AID dataset. Key findings include:

1. **EfficientNet-B0** achieves the strongest linear probe transfer (91.14%) and full fine-tuning accuracy (95.29%) despite being the smallest model (4.05 M parameters), with the highest feature separability ratio (0.8090).
2. **EfficientNet-B0 is the most robust architecture overall** under distribution shift (mean RR ≈ 0.41), leading under Gaussian noise (RR = 0.40) and brightness shift (RR = 0.49). ResNet-50 leads under motion blur (RR = 0.59).
3. **DenseNet-121** is the most data-efficient backbone ($\Delta = 22.71\%$ few-shot drop), but the least robust under Gaussian noise (RR = 0.096 at $\sigma = 0.20$).
4. **Layer-wise probing** confirms that late layers encode the most discriminative semantics (92–94% probing accuracy), while mid-layers offer the best robustness-accuracy trade-off for transfer under distribution shift.
5. **Selective 20% unfreezing** achieves within 1–3% of the best fine-tuning strategy while updating only $\sim 20\%$ of parameters, validating the automated deep-layer selection hypothesis.
6. **Gaussian noise** is the most damaging corruption across all models, with all architectures losing $>55\%$ of clean accuracy, highlighting the brittleness of standard CNN training without robustness augmentation.